

Relational schema

CIS 5450, Fall 2026

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Engineering
UNIVERSITY of PENNSYLVANIA

Outline

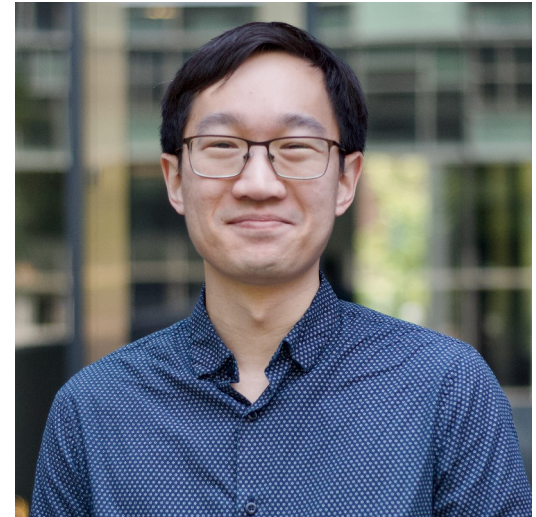
- Course logistics
- Brief review
- Relational schema
- Understanding schema with ER diagrams
- Understanding schema with normal forms

Outline

- **Course logistics**
- Brief review
- Relational schema
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- Understanding schema with normal forms

Upcoming Deadlines

- HW1 due TODAY @ 10pm
- HW2 released at 10pm, due in 2 weeks
- Next class, guest lecture from Jeff Tao



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Combining Data

- Combining data **that was intended to be combined**
- Combining data that was **not** intended to be combined

Joining

- Pure computation

Data Integration

- More complex
- Requires knowledge

Single Source of Truth

- Joins may seem inconvenient, but can greatly increase data quality.

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

Redundant data!

What happens if the CIS admin changes?

- Have to update **multiple cells**
 - Bad for performance
 - Error-prone

Joins in Pandas

df1

	a	b
0	1	alpha
1	2	beta
2	3	gamma

df2

	c	d
0	1	NM
1	1	RI
2	2	PA



```
df1.merge(df2, left_on="a", right_on="c")
```

	a	b	c	d
0	1	alpha	1	NM
1	1	alpha	1	RI
2	2	beta	2	PA

Joins in Pandas

- How many movies starring Angelina Jolie were produced by Sony?

company_name

id
name

title

id
title

name

id
name

movie_companies

company_id
movie_id

cast_info

person_id

Entity tables

Joins in Pandas

- How many movies starring Angelina Jolie were produced by Sony?

company_name

id
name

title

id
title

name

id
name

Links
companies
to movies

movie_companies

company_id
movie_id

cast_info

movie_id
person_id

Links
actors to
movies

Joins in Pandas

- How many movies starring Angelina Jolie were produced by Sony?

company_name

title

name

id
name

Relationship tables

movie_companies

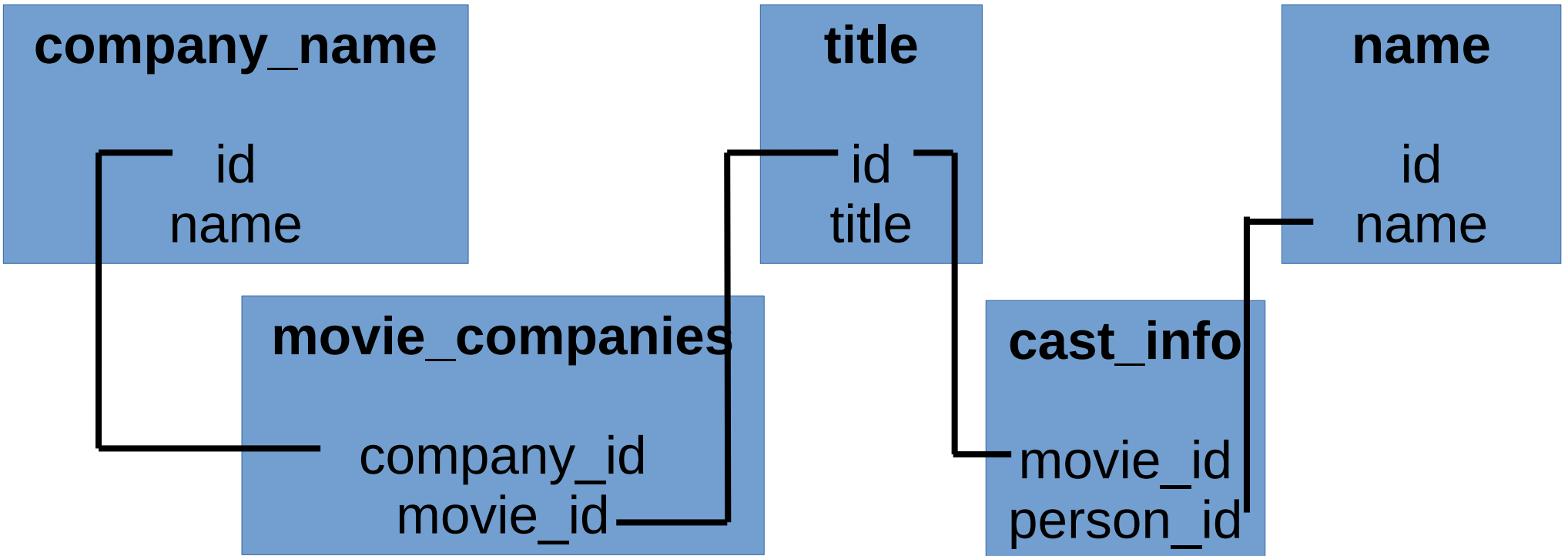
company_id
movie_id

cast_info

movie_id
person_id

Joins in Pandas

- How many movies starring Angelina Jolie were produced by Sony?

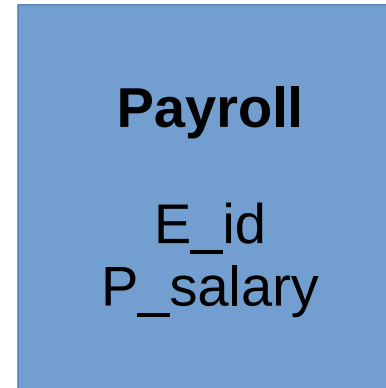
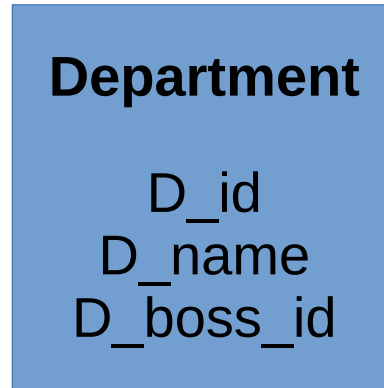
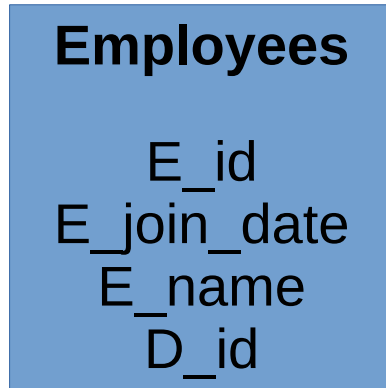


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- **Relational schema**
- Understanding schema with ER diagrams
- Understanding schema with normal forms

Relational schema

- “Relational” = tabular = tables = dataframes
- Relational **schema** is what the tables and columns represent.



Some exceptions apply

Relational schema in SQL

Employees	
e_id	
e_join_date	
e_name	
d_id	

Relation

```
CREATE TABLE Employee (  
  e_id INT,  
  e_join_date TIMESTAMP,  
  e_name TEXT,  
  d_id INT,  
)
```

Columns / attributes

Data types

Datatypes

Each value stored in an SQLite database (or manipulated by the database engine) has one of the following storage classes:

- **NULL**. The value is a NULL value.
- **INTEGER**. The value is a signed integer, stored in 0, 1, 2, 3, 4, 6, or 8 bytes depending on the magnitude of the value.
- **REAL**. The value is a floating point value, stored as an 8-byte IEEE floating point number.
- **TEXT**. The value is a text string, stored using the database encoding (UTF-8, UTF-16BE or UTF-16LE).
- **BLOB**. The value is a blob of data, stored exactly as it was input.

SQLite

Datatypes

Each value stored in an SQLite database (or manipulated by the database engine) has one of the following storage classes:

- **NULL**. The value is a NULL value.
- **INTEGER**. The value is a signed integer, stored in a 2, 4, or 8 byte integer, depending on the magnitude of the value.
- **REAL**. The value is a floating point value, stored as a double-precision floating point number.
- **TEXT**. The value is a text string, stored using UTF-8, UTF-16BE, or UTF-16LE.
- **BLOB**. The value is a blob of data, stored as a blob.

Name	Aliases	Description
BIGINT	INT8, LONG	signed eight-byte integer
BIT	BITSTRING	string of 1s and 0s
BLOB	BYTEA, BINARY, VARBINARY	variable-length binary data
BOOLEAN	BOOL, LOGICAL	logical boolean (true/false)
DATE		calendar date (year, month day)
DECIMAL(prec, scale)	NUMERIC(prec, scale)	fixed-precision number with the given width (precision) and scale, defaults to prec = 18 and scale = 3
DOUBLE	FLOAT8,	double precision floating-point number (8 bytes)
HUGEINT		signed sixteen-byte integer
INTEGER	INT4, INT, SIGNED	signed four-byte integer
INTERVAL		date / time delta
REAL	FLOAT4, FLOAT	single precision floating-point number (4 bytes)
SMALLINT	INT2, SHORT	signed two-byte integer
TIME		time of day (no time zone)
TIMESTAMP WITH TIME ZONE	TIMESTAMPZ	combination of time and date that uses the current time zone
TIMESTAMP	DATETIME	combination of time and date
TINYINT	INT1	signed one-byte integer
UBIGINT		unsigned eight-byte integer
UHUGEINT		unsigned sixteen-byte integer
UINTEGER		unsigned four-byte integer
USMALLINT		unsigned two-byte integer
UTINYINT		unsigned one-byte integer
UUID		UUID data type

Primary and foreign keys

Primary key: a unique identifier for each row of the table.

Foreign key: a non-unique reference to a unique column of another table.

Employees

e_id
e_join_date
e_name
d_id

```
CREATE TABLE Employee (  
    e_id INT,  
    e_join_date TIMESTAMP,  
    e_name TEXT,  
    d_id INT,  
    PRIMARY KEY (e_id),  
    FOREIGN KEY (d_id) REFERENCES Department(d_id)  
)
```

Relational schema in SQL

- Unique key groups
- Checked constraints
- Custom data types
- External tables

Why do we have schemas?

- Schemas ensure all of our data follows the same rules
 - Benefit 1: helps keep data clean (*integrity*)
 - Benefit 2: makes computation easier
 - Benefit 3: provides a “map” of the data that is guaranteed to be true

Outline

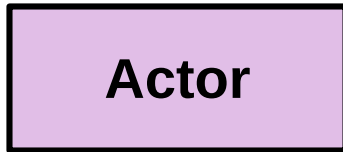
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- **Understanding schema with ER diagrams**
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ER Diagrams

- ER (entity-relationship) diagrams are a common tool to *design* and **document** databases.
- Visual representation of the “nouns” and “verbs” inside of a schema.

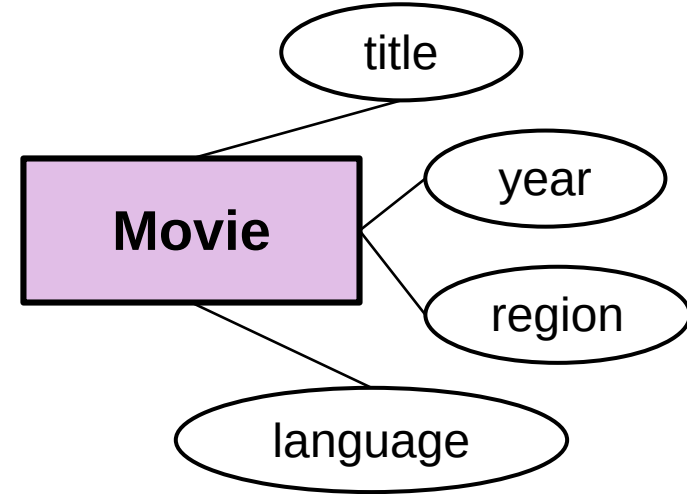
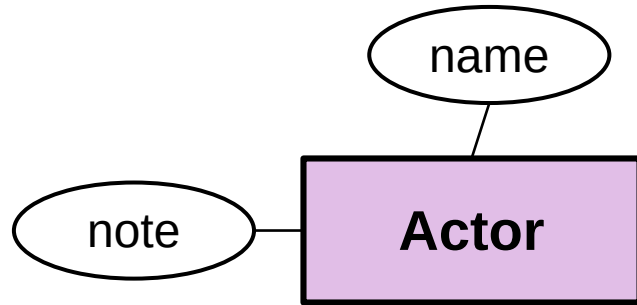
ER: Entities

- *Entities* are the *objects* of the schema (rectangles)



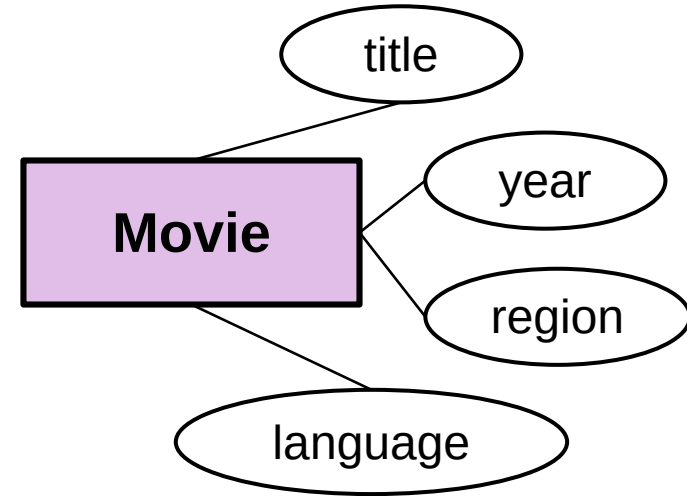
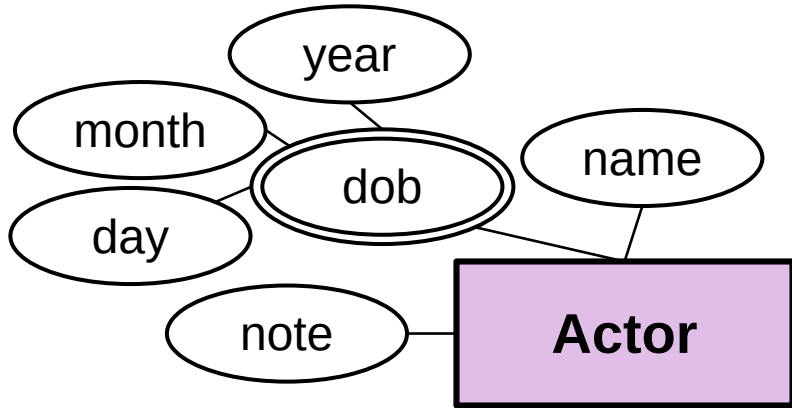
ER: Entities

- *Attributes* are things that describe entities (ovals)



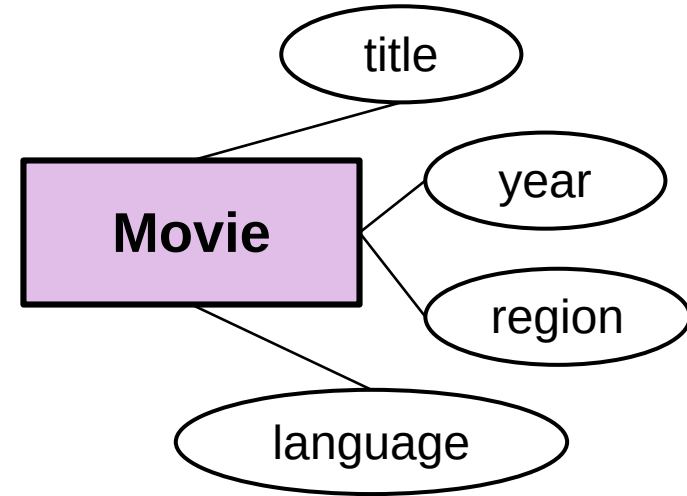
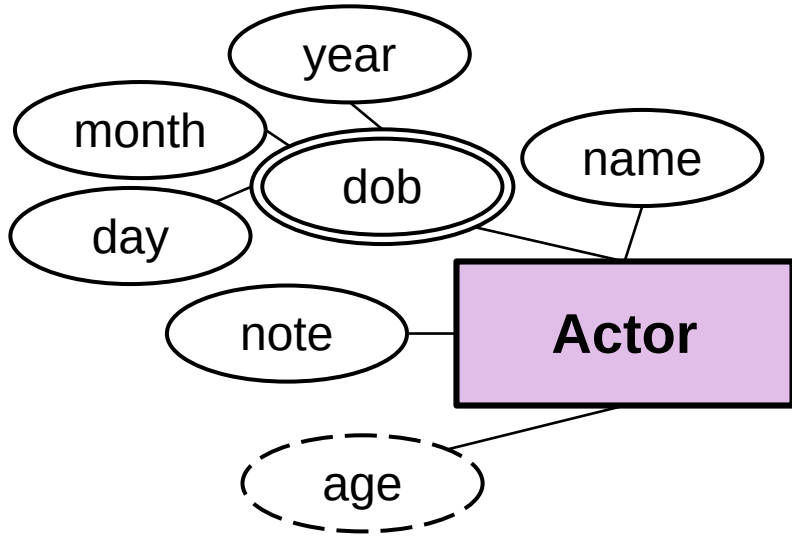
ER: Entities

- *Compound attributes* are made up of other attributes (double oval)



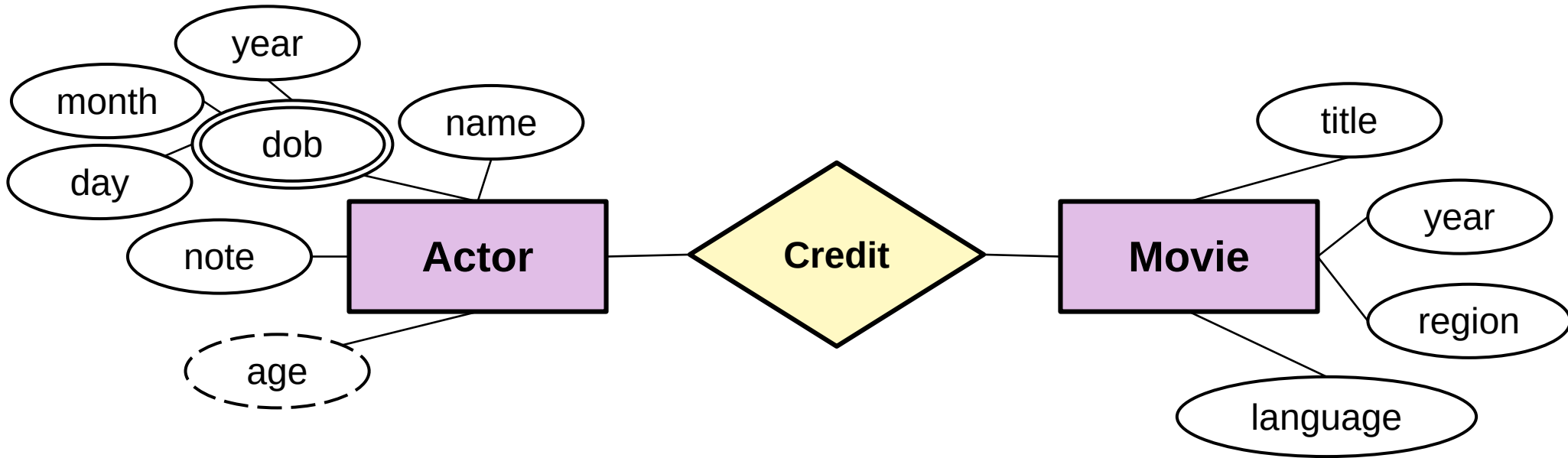
ER: Entities

- *Computed attributes* are attributes that can be computed from other attributes (dashed oval)



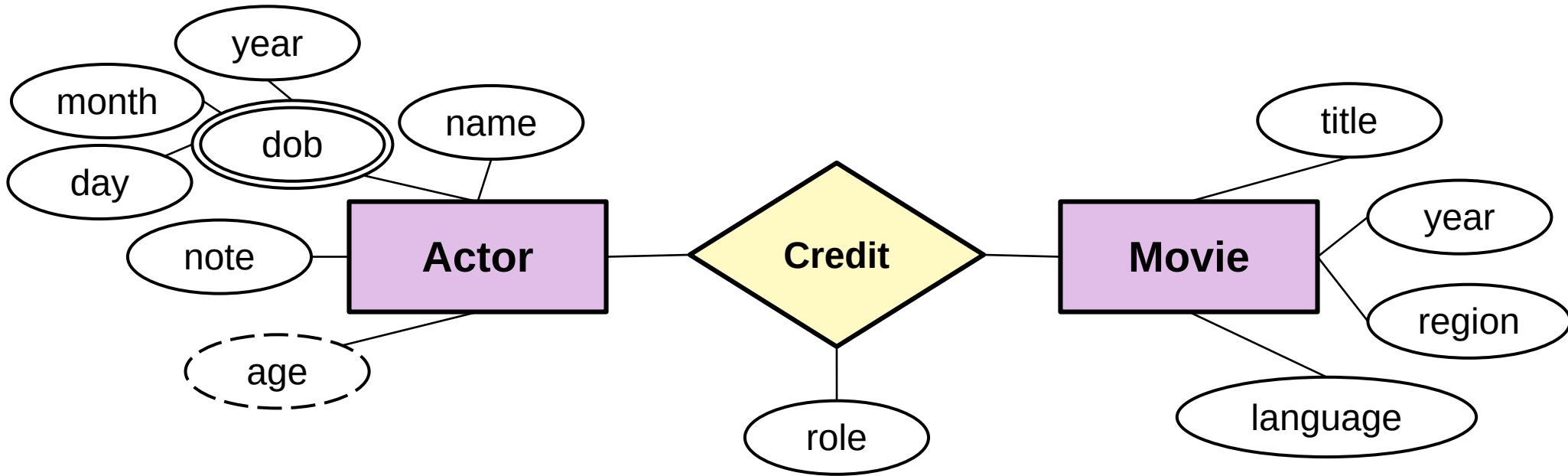
ER: Entities

- *Relationships* are the *verbs* of the schema (diamonds)



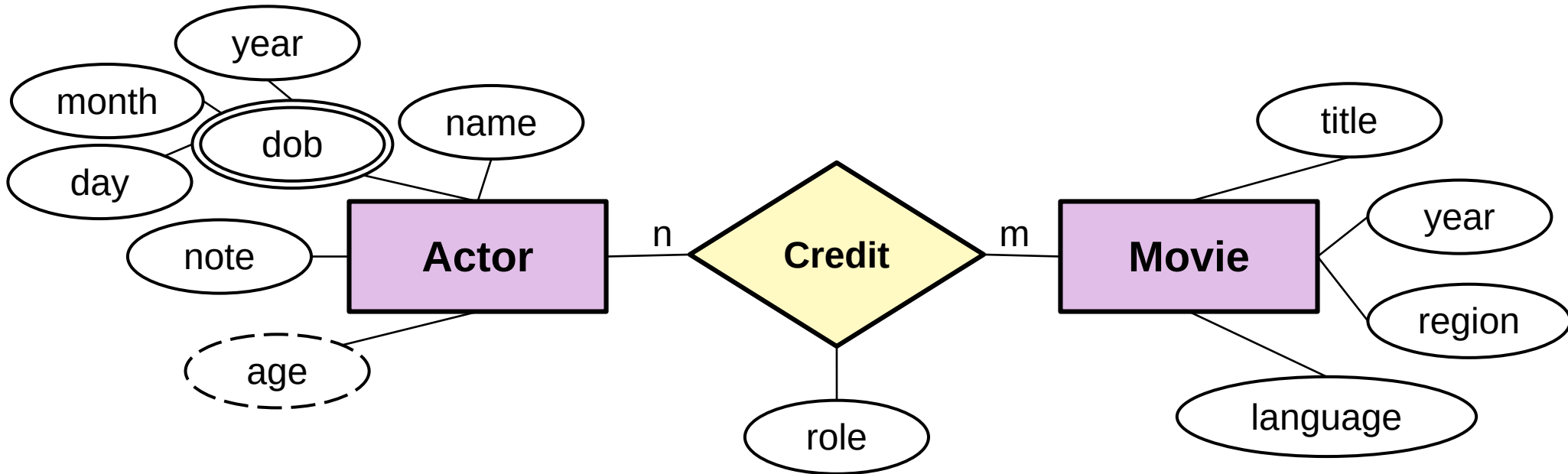
ER: Entities

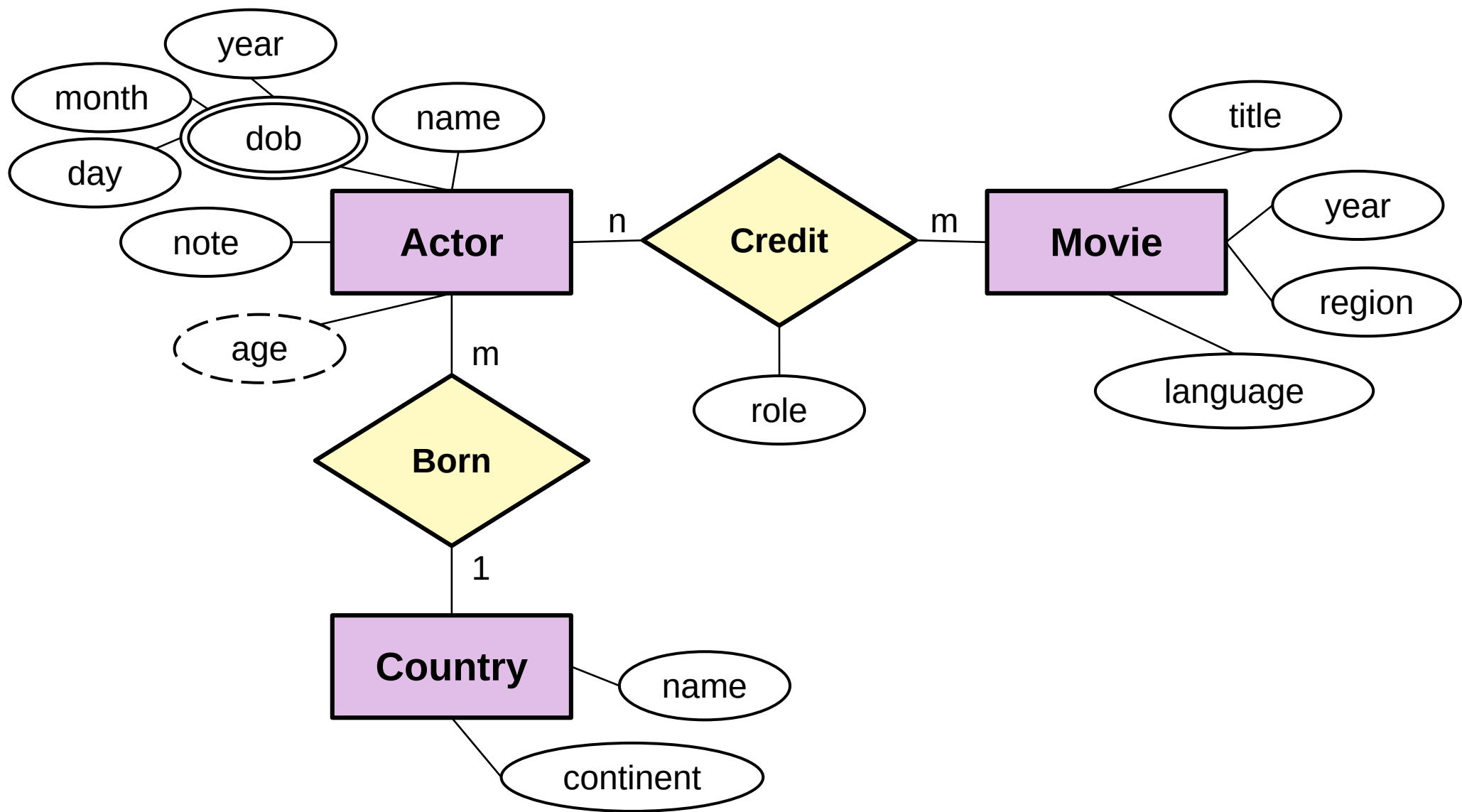
- Relationships can have attributes too!

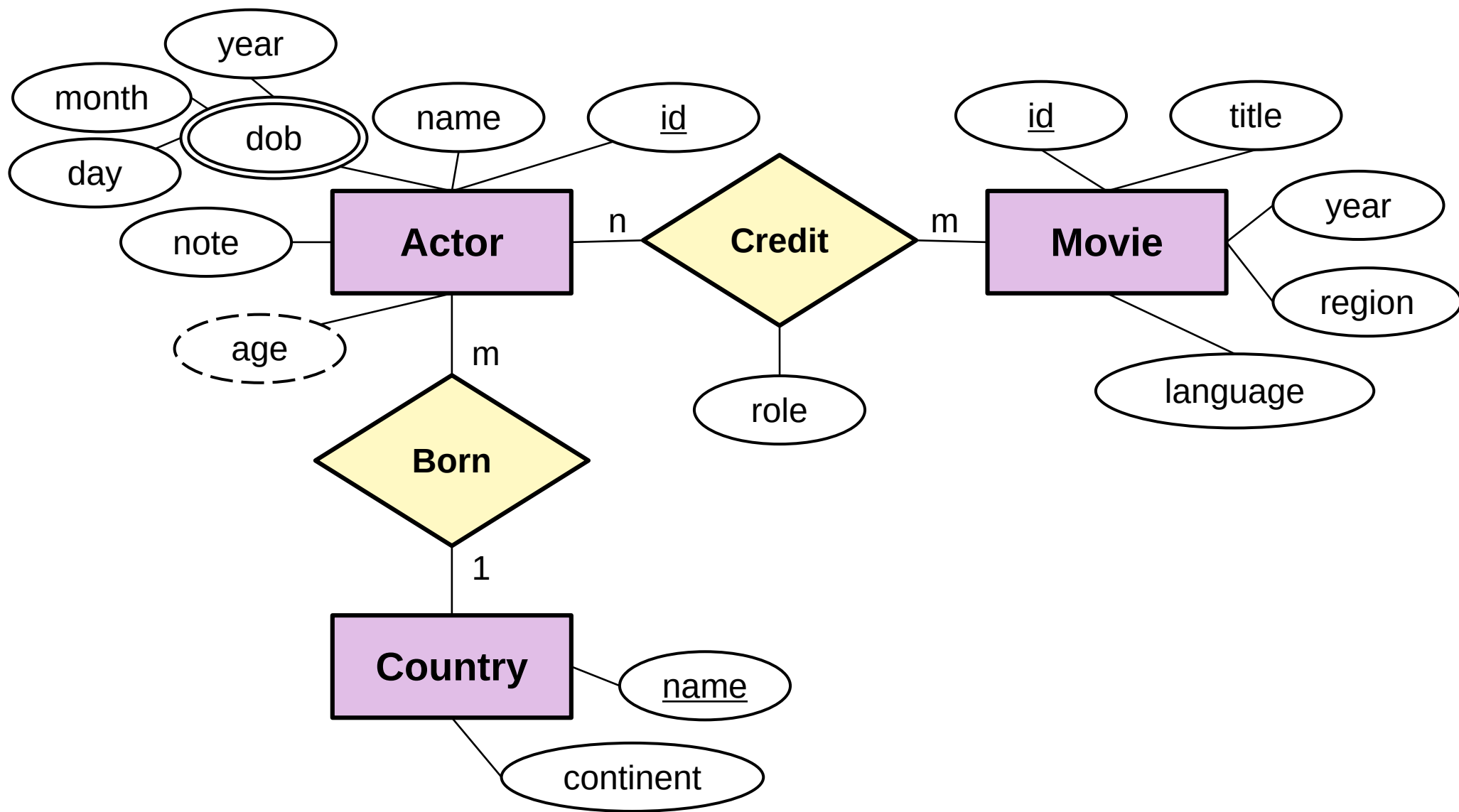


ER: Entities

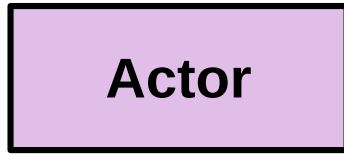
- Relationships have a *degree*







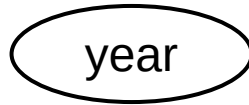
ER Diagrams



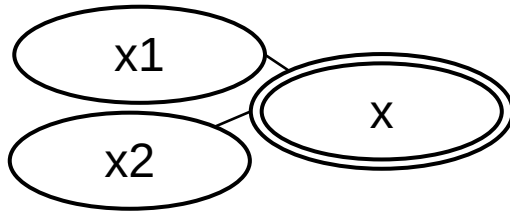
Entities; the objects or nouns of the schema



Relationships; the verbs or linkages of the schema



Attributes, descriptors of entities or relationships



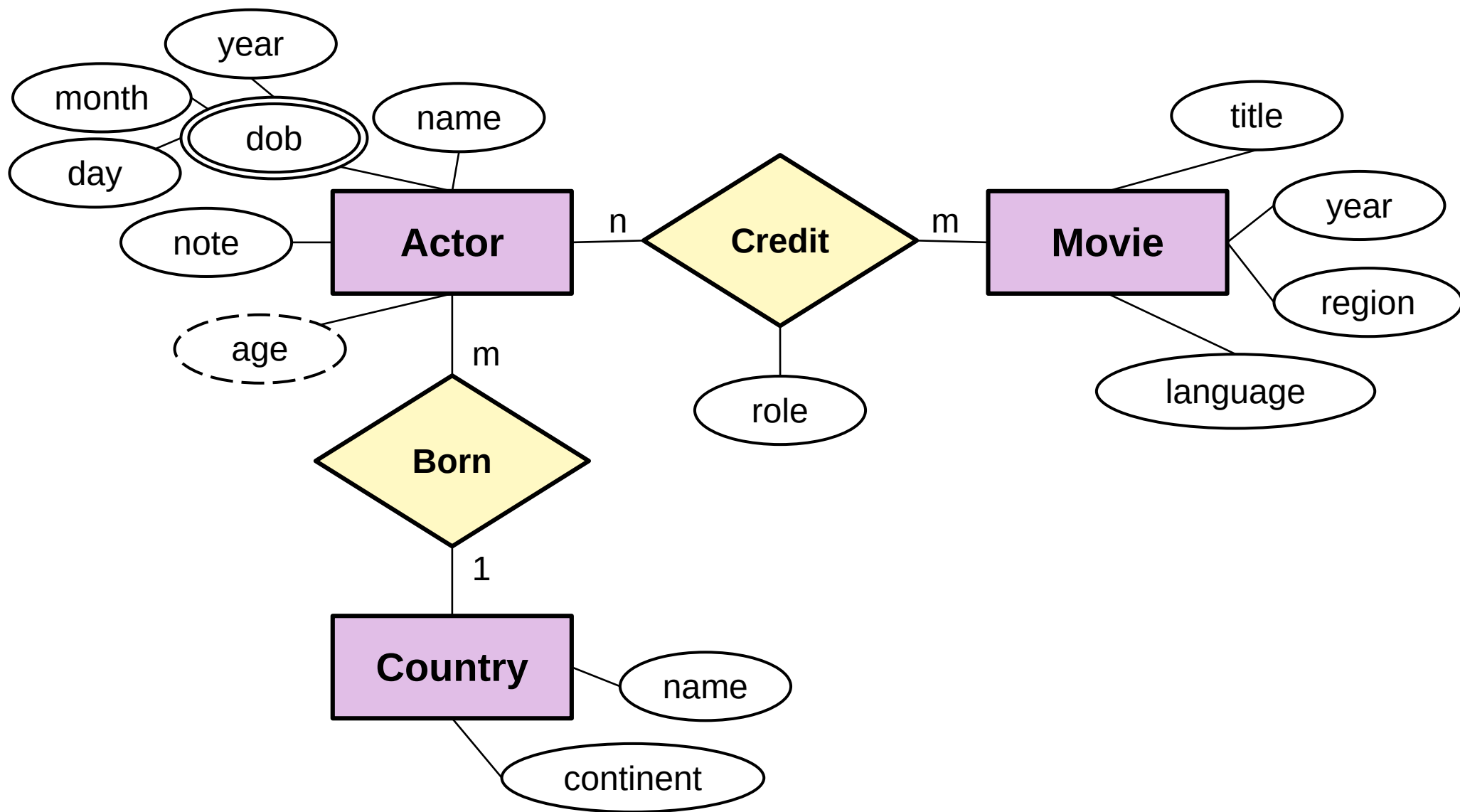
Compound attributes are composed of other attributes



Computed attributes can be determined from other attrs

From ER to Schema

- Given an ER diagram, how to get a schema?
- Each *entity* becomes a table, each attribute a column
 - Compound and computed attributes are special cases
- Each relationship becomes either:
 - A table with two foreign keys (many-to-many)
 - A foreign key in another table (one-to-many)



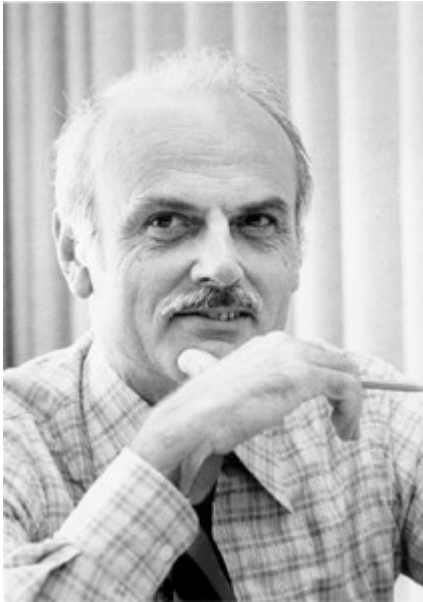
ER Diagrams

- Advantages:
 - Visual
 - Easy to understand
 - Clear correspondence to the real world
- Disadvantages:
 - Imprecise: how do we know we have the right # of tables?
 - Subjective: difficult to decide if an ER diagram is “right” or “wrong”
 - Is Country related to Continent or is Continent an attribute?

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First Normal Form - 1NF



Edgar F. Codd

- Today we will learn 1NF, 2NF, 3NF.
 - Each applies to a table, such that a table is either “in” a normal form or “violating” a normal form.
 - Each is defined in terms of the previous form.

First Normal Form - 1NF

Customer	Customer ID	Transactions		
Abraham	1	Transaction ID	Date	Amount
		12890	2003-10-14	-87
		12904	2003-10-15	-50
Isaac	2	Transaction ID	Date	Amount
		12898	2003-10-14	-21
Jacob	3	Transaction ID	Date	Amount
		12907	2003-10-15	-18
		14920	2003-11-20	-70
		15003	2003-11-27	-60

- Today we will learn 1NF, 2NF, 3NF.
 - Each applies to a table, such that a table is either “in” a normal form or “violating” a normal form.
 - Each is defined in terms of the previous form.
- 1NF is the simplest: all values are themselves not tables

First Normal Form - 1NF

Customer	Customer ID	Transactions		
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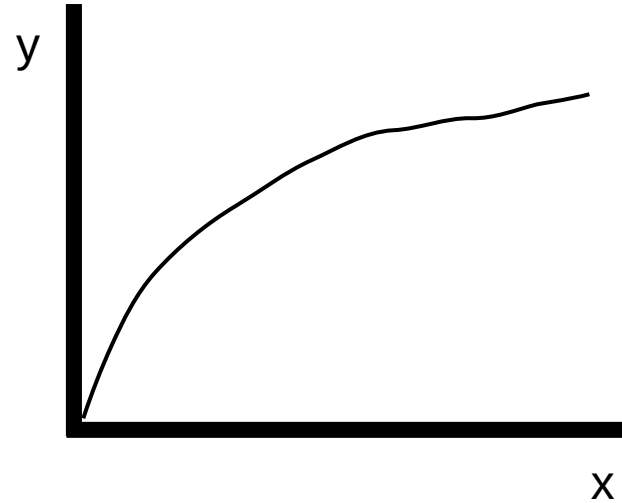
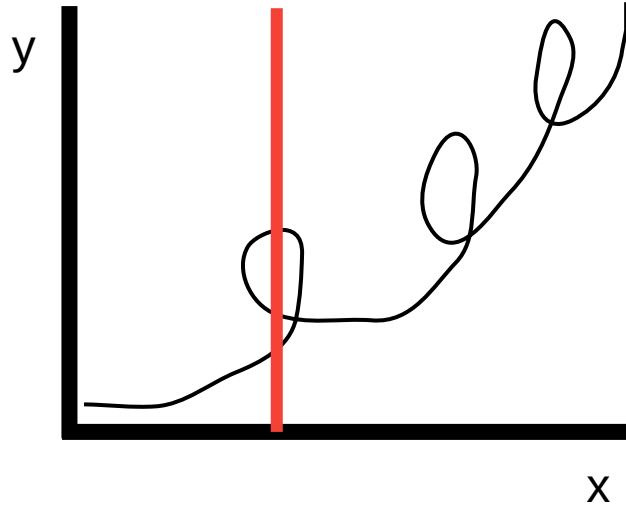
Cust	CustID	T_ID	Date	Amt
Abraham	1	12890	2003-10-14	-87
Abraham	1	12904	2003-10-15	-50
Isaac	2	12898	2003-10-14	-21
Jacob	3	12907	2003-10-15	-18
Jacob	3	14920	2003-11-20	-70
Jacob	3	15003	2003-11-27	-60

previous form.

- 1NF is the simplest: all values are themselves not tables

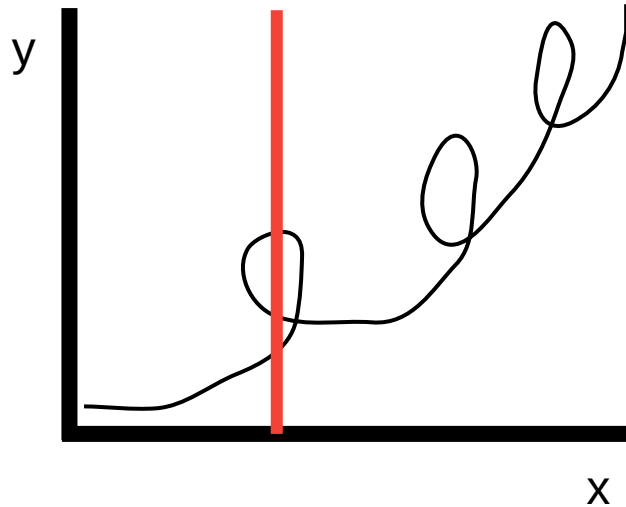
Functional Dependencies

- Recall: the “vertical line test”

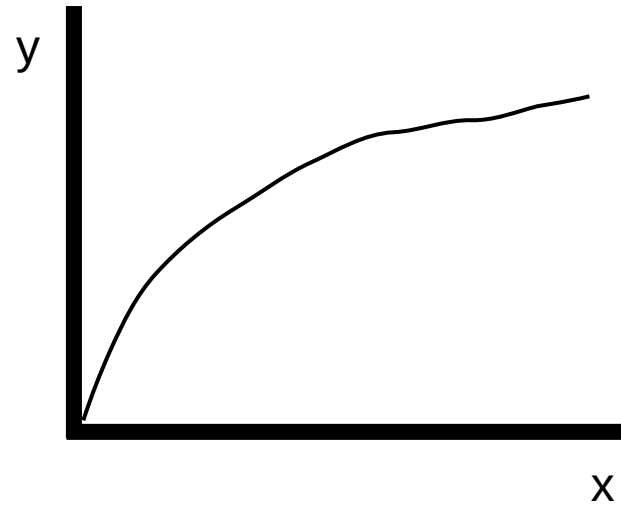


Functional Dependencies

- Recall: the “vertical line test”



Not a function



A function

Functional Dependencies

- More precisely:
 - Say $f(x)$ is a function iff for all x , $f(x)$ has only 1 value
 - Or:
 - Given $f(x_1) = y_1$ and $f(x_2) = y_2$,
 - And given $y_1 \neq y_2$,
 - Then: x_1 must not equal x_2

Ex:

If $f(x_1) = 5$ and $f(x_2) = 7$

Then EITHER:
 $x_1 \neq x_2$ OR $f(x)$ is not a function
- We say a table has a functional dependency between attributes A and B , written $A \rightarrow B$, iff B is a function of A .

$A \rightarrow B$ read aloud is “ A determines B ”

Functional Dependencies

if $f(x_1) = y_1$ and $f(x_2) = y_2$, then if $y_1 \neq y_2$, x_1 must not equal x_2

Does Dept $\rightarrow$ Professor?

In other words: is there a function $f(\text{department}) \rightarrow \text{professor}$?

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

Functional Dependencies

if $f(x_1) = y_1$ and $f(x_2) = y_2$, then if $y_1 \neq y_2$, x_1 must not equal x_2

Does Dept $\rightarrow$ Professor? **No**. Proof:

$X_1 = \text{CIS}$ $Y_1 = \text{Marcus}$

$X_2 = \text{CIS}$ $Y_2 = \text{Gardner}$

$Y_1 \neq Y_2$, but $X_1 = X_2$ – no function!

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

Functional Dependencies

if $f(x_1) = y_1$ and $f(x_2) = y_2$, then if $y_1 \neq y_2$, x_1 must not equal x_2

Does Dept $\rightarrow$ Admin?

Professor	Dept	Admin
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Functional Dependencies

if $f(x_1) = y_1$ and $f(x_2) = y_2$, then if $y_1 \neq y_2$, x_1 must not equal x_2

Does Dept $\rightarrow$ Admin? **Yes**. Proof:

X1 = CIS Y1 = Mari
X2 = CIS Y2 = Mari

X1 = CIS Y1 = Mari
X2 = COM Y2 = Andy

Check every pair of rows – no violation.

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

Functional Dependencies

if $f(x_1) = y_1$ and $f(x_2) = y_2$, then if $y_1 \neq y_2$, x_1 must not equal x_2

Does Prof $\rightarrow$ Dept?

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Functional Dependencies

if $f(x_1) = y_1$ and $f(x_2) = y_2$, then if $y_1 \neq y_2$, x_1 must not equal x_2

Does Prof $\rightarrow$ Dept? **Yes**. Proof:

X1 = Marcus Y1 = CIS X1 = Marcus Y1 = CIS X1 = Gardner Y1 = CIS
X2 = Gardner Y2 = CIS X2 = Gerbner Y2 = COM X2 = Gerbner Y2 = COM

Check every pair of rows – no violation.

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

Functional Dependencies

- Intuitive way to check for Fds:
 - Say that $A \rightarrow B$ if, given A, you can always say the exact value of B.
- Given a professor, can you say what dept they are in?
 $\text{Prof} \rightarrow \text{Dept}$ Yes!
- Given a department, can you tell me the department's admin?
 $\text{Dept} \rightarrow \text{Admin}$ Yes!
- Given an admin, can you tell me the professor?
 $\text{Admin} \rightarrow \text{Prof}$ No!

Algebra of FDs

- Armstrong's axioms
- Reflexivity: Y is a subset of X , then $X \rightarrow Y$
 - Ex: the prof and the dept determine the dept
- Augmentation: if $X \rightarrow Y$, then for all Z , $XZ \rightarrow YZ$
 - Ex: $\text{prof} \rightarrow \text{dept}$, so $(\text{prof}, \text{admin}) \rightarrow (\text{dept}, \text{admin})$
- Transitivity: if $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$
 - Ex: $\text{prof} \rightarrow \text{dept}$ and $\text{dept} \rightarrow \text{admin}$, so $\text{prof} \rightarrow \text{admin}$

These axioms are **complete** and **sound**

Proving more properties

- Decomposition:
 - if $X \rightarrow YZ$, then $X \rightarrow Y$

Reflex: Y in X , $X \rightarrow Y$

Augm: $X \rightarrow Y$, for any Z , $XZ \rightarrow YZ$

Trans: $X \rightarrow Y$ and $Y \rightarrow Z$, $X \rightarrow Z$

Proof:

Step 1: $X \rightarrow YZ$ (given)

Step 2: $YZ \rightarrow Y$ (reflex)

Step 3: $X \rightarrow Y$ (trans of 1 + 2)

Proving more properties

- Union:
 - if $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$

Proof:

Step 1: $X \rightarrow Y$ (given)
Step 2: $X \rightarrow Z$ (given)
Step 3: $X \rightarrow XZ$ (augm 2 with X)
Step 4: $XZ \rightarrow YZ$ (augm 1 with Z)
Step 5: $X \rightarrow YZ$ (trans 3 + 4)

Reflex: Y in X , $X \rightarrow Y$

Augm: $X \rightarrow Y$, for any Z , $XZ \rightarrow YZ$

Trans: $X \rightarrow Y$ and $Y \rightarrow Z$, $X \rightarrow Z$

Proving more properties

- Prove: if $X \rightarrow Y$ and $YZ \rightarrow W$, then $XZ \rightarrow W$

Reflex: Y in X , $X \rightarrow Y$

Augm: $X \rightarrow Y$, for any Z , $XZ \rightarrow YZ$

Trans: $X \rightarrow Y$ and $Y \rightarrow Z$, $X \rightarrow Z$

Proving more properties

- Prove: if $X \rightarrow Y$ and $YZ \rightarrow W$, then $XZ \rightarrow W$

Proof:

Step 1: $X \rightarrow Y$ (given)
Step 2: $YZ \rightarrow W$ (given)
Step 3: $XZ \rightarrow YZ$ (augm 1 with Z)
Step 4: $XZ \rightarrow W$ (trans 3 and 2)

Reflex: Y in X , $X \rightarrow Y$

Augm: $X \rightarrow Y$, for any Z , $XZ \rightarrow YZ$

Trans: $X \rightarrow Y$ and $Y \rightarrow Z$, $X \rightarrow Z$

Lossless Joins

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

Professor	Dept		
Marcus	CIS		
Gardner	CIS	Dept	Admin
Gerbner	COM	CIS	Mari
		COM	Andy

Good!

Professor	Admin		
Marcus	Mari		
Gardner	Mari	Dept	Admin
Gerbner	Andy	CIS	Mari
		CIS	Mari
		COM	Andy

... bad!

Lossless Joins

- When can we turn one relation R into R_1 and R_2 ?
 - When the join between them is lossless (reconstructs the original table)
- Theorem (*lossless join decomposition*):
 - (R_1, R_2) is a lossless decomposition of R iff
 - $R_1 \cap R_2 \rightarrow R_2$

Lossless Joins

Theorem (*lossless join decomposition*):

- Given $R_1 \cup R_2 = R$, when does $R_1 \bowtie R_2 = R$?
- (R_1, R_2) is a lossless decomposition of R iff
- $R_1 \cap R_2 \rightarrow R_2$

$R = (\text{Prof, Dept, Admin})$

$R_1 = (\text{Prof, Dept})$

$R_2 = (\text{Dept, Admin})$

$R_1 \cap R_2 = (\text{Dept})$

Does $\text{Dept} \rightarrow \text{Dept, Admin}$?

Yes, by decomposition

Lossless Joins

Theorem (*lossless join decomposition*):

- Given $R_1 \cup R_2 = R$, when does $R_1 \bowtie R_2 = R$?
- (R_1, R_2) is a lossless decomposition of R iff
- $R_1 \cap R_2 \rightarrow R_2$

$R = (\text{Prof}, \text{Prof_Sal}, \text{Dept}, \text{Admin})$

$R_1 = (\text{Prof}, \text{Dept})$

$R_2 = (\text{Dept}, \text{Admin}, \text{Prof_Sal})$

$R_1 \cap R_2 = (\text{Dept})$

Does $\text{Dept} \rightarrow \text{Dept}, \text{Admin}, \text{Prof_Sal}$?

No.

Keys

- Now that we know *when* a decomposition is lossless, now we need to figure out which decomposition to pick!
- Next, we need to understand *keys*.

Keys

$\{ABCDE\}$ and $\{ABCD\}$
are **super keys**

$\{ABD\}$ and $\{ACD\}$ are
candidate keys

$\{ABD\}$ can be picked
as **primary key**

- Suppose $R = \{A, B, C, D, E\}$
 - $AB \rightarrow CE, CD \rightarrow BE$
- Any set of attributes that determines R is a *superkey*
- If a superkey is *minimal*, we call it a *candidate key*
- We pick one candidate key to be our *primary key*

Keys

{ABCDE} and {ABCD}
are **super keys**

{ABD} and {ACD} are
candidate keys

{ABD} can be picked
as **primary key**

- Suppose $R = \{\underline{A}, \underline{B}, \underline{C}, \underline{D}, E\}$
 - $AB \rightarrow CE, CD \rightarrow BE$
- An attribute that is part of *any* candidate key is a **prime attribute**
- If a *candidate key* is not a *primary key*, we call it a *candidate key*
- We pick one candidate key to be our *primary key*

Second Normal Form

- R is in 2NF iff: R is in 1NF and no non-prime attribute is determined by any proper subset of a candidate key.

$R = \{\underline{A}, \underline{B}, \underline{C}, \underline{D}, E\}$
 $AB \rightarrow CE, CD \rightarrow BE$

{ABD} and {ACD} are
candidate keys

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

$P \rightarrow D, D \rightarrow A$
Candidate keys: {P}

Violates 2NF! $AB \rightarrow E$

All good!

Second Normal Form

$R = \{\underline{A}, \underline{B}, \underline{C}, \underline{D}, E\}$

$AB \rightarrow CE, CD \rightarrow BE$

$\{ABD\}$ and $\{ACD\}$ are
candidate keys

Violates 2NF! $AB \rightarrow E$

$R_1 = R - \text{RHS of FD}$

$R_1 = \{A, B, C, D\}$

$AB \rightarrow C$

$CD \rightarrow B$

CKs: $\{ABD\} \{ACD\}$

$R_2 = \text{LHS} + \text{RHS of FD}$

$R_2 = \{A, B, E\}$

$AB \rightarrow E$

CKs: $\{AB\}$

Second Normal Form

$R = \{\underline{A}, \underline{B}, \underline{C}, \underline{D}, E\}$
 $AB \rightarrow CE, CD \rightarrow BE$

$\{ABD\}$ and $\{ACD\}$ are
candidate keys

Violates 2NF! $AB \rightarrow E$

R is in 2NF iff: no non-prime attribute is determined by any proper subset of a candidate key.

$R_1 = R - \text{RHS of FD}$

$R_1 = \{\underline{A}, \underline{B}, \underline{C}, \underline{D}\}$

$AB \rightarrow C$
 $CD \rightarrow B$

CKs: $\{ABD\} \{ACD\}$

All good!

$R_2 = \text{LHS} + \text{RHS of FD}$

$R_2 = \{\underline{A}, \underline{B}, E\}$

$AB \rightarrow E$

CKs: $\{AB\}$

All good!

Second Normal Form

If I always use this criteria to pick a decomposition, will I always get a lossless join?

Recall that a join is lossless if:

$$R_1 \cap R_2 \rightarrow R_2$$

$R_1 = R - \text{RHS of FD}$

$R_1 = \{\underline{\mathbf{A}}, \underline{\mathbf{B}}, \underline{\mathbf{C}}, \underline{\mathbf{D}}\}$

$AB \rightarrow C$

$CD \rightarrow B$

CKs: {ABD} {ACD}

$R_2 = \text{LHS} + \text{RHS of FD}$

$R_2 = \{\underline{\mathbf{A}}, \underline{\mathbf{B}}, \mathbf{E}\}$

$AB \rightarrow E$

CKs: {AB}

Third Normal Form

- R is in 1NF if every value is a value
- R is in 2NF if:
 - R is in 1NF AND
 - R is in 1NF and no non-prime attribute is determined by any proper subset of a candidate key.
- R is in 3NF if:
 - R is in 2NF AND
 - No non-prime attribute determines a different non-prime attribute

Third Normal Form

No non-prime
attribute determines a
different non-prime
attribute

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

$P \rightarrow D, D \rightarrow A$

Candidate keys: {P}

Third Normal Form

No non-prime
attribute determines a
different non-prime
attribute

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

$P \rightarrow D, D \rightarrow A$

Candidate keys: {P}

Third Normal Form

No non-prime attribute determines a different non-prime attribute

Professor	Dept	Admin
Marcus	CIS	Mari
Gardner	CIS	Mari
Gerbner	COM	Andy

$R_1 = R - \text{RHS of FD}$
 $R_1 = \{\text{Prof}, \text{Dept}\}$

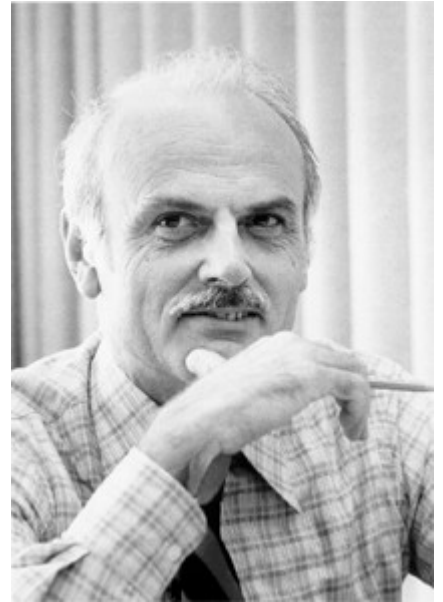
$P \rightarrow D, D \rightarrow A$
Candidate keys: $\{P\}$

$R_2 = \text{LHS} + \text{RHS of FD}$
 $R_2 = \{\text{Dept}, \text{Admin}\}$

Third Normal Form

Every non-prime attribute is determined by...

- The keys (by definition)
- The whole keys (2NF)
- ... and nothing but the keys (3NF)



Edgar F. Codd

The 3NF Algorithm

Repeat until all relations are in 3NF:

- 1) Pick a relation we haven't checked yet.
- 2) Ensure the relation is in 1NF.
- 3) Write down the FDs
- 4) Identify the CKs
- 5) Check for 2NF violations
 - If any, decompose on a violating FD, go to (1)
- 6) Check for 3NF violations
 - If any, decompose on a violating FD, go to (1)

ActorID	Actor Name	ActorDOB	Country Name	Country Continent	Movies
1	Jolie, Angeli	June 4 th , 1975	US	NA	[{"role": "Lara Croft", "movieID": 1, "movieName": "Tomb Raider"}, ...]
2	Li, Jet	April 26 th , 1963	CN	AS	[{"role": "Huo", "movieID": 2, "movieName": "Fearless"}, {"role": "Liu", "movieID": 3, "movieName": "Kiss of the Dragon"}, ...]

ActorID	Actor Name	ActorDOB	Country Name	Country Continent	Movies
1	Jolie, Angeli	June 4 th , 1975	US	NA	[{"role": "Lara Croft", "movieID": 1, "movieName": "Tomb Raider"}, ...]
2	Li, Jet	April 26 th , 1963	CN	AS	[{"role": "Huo", "movieID": 2, "movieName": "Fearless"}, {"role": "Liu", "movieID": 3, "movieName": "Kiss of the Dragon"}, ...]

Step 1:
Not in 1NF! Table-in-a-table.

Actor ID	Actor Name	ActorDOB	Country Name	Country Cont	Role	Movie ID	Movie Name
1	Jolie, Angeli	June 4 th , 1975	US	NA	Lara	1	Tomb Raider
1	Jolie, Angeli	June 4 th , 1975	US	NA	Margaret	4	The Good Shepherd
2	Li, Jet	April 26 th , 1963	CN	AS	Huo	2	Fearless
2	Li, Jet	April 26 th , 1963	CN	AS	Liu	3	Kiss of the Dragon

Step 2: Write down our FDs:

ActorID → ActorName, ActorDOB, CountryName, CountryCont

CountryName → CountryCont

MovieID → MovieName

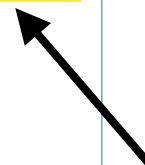
Step 3: Find candidate keys (minimal keys)

CKs: {ActorID, MovieID, Role}

Step 4: Check for 2NF

Are any non-prime attributes determined by a subset of a candidate key?

Violation!



$R = \{ActorID, ActorName, ActorDOB, CountryName, CountryCont, Role, MovieID, MovieName\}$

Step 2: Write down our FDs:

$ActorID \rightarrow ActorName, ActorDOB, CountryName, CountryCont$

$CountryName \rightarrow CountryCont$

$MovieID \rightarrow MovieName$

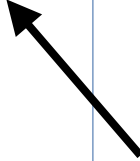
Step 3: Find candidate keys (minimal keys)

CKs: $\{ActorID, MovieID, Role\}$

Step 4: Check for 2NF

Are any non-prime attributes determined by a subset of a candidate key?

Violation!



Step 5: Decompose with the violating FD

$R_1 = R - \text{RHS of FD}$

$R_1 = \{ActorID, Role, MovieID, MovieName\}$

$R_2 = \text{LHS} + \text{RHS of FD}$

$R_2 = \{ActorID, ActorName, ActorDOB, CountryName, CountryCont\}$

$R_1 = \{\text{ActorID}, \text{Role}, \text{MovieID}, \text{MovieName}\}$

$R_2 = \{\text{ActorID}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}, \text{CountryCont}\}$

$R_1 = \{\text{ActorID}, \text{Role}, \text{MovieID}, \text{MovieName}\}$

$R_2 = \{\text{ActorID}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}, \text{CountryCont}\}$

Step 1: Check for 1NF
... generally always good

Step 2: Write down our FDs:

$\text{MovieID} \rightarrow \text{MovieName}$

Step 3: Find candidate keys (minimal keys)

CKs: $\{\text{ActorID}, \text{Role}, \text{MovieID}\}$

Step 4: Check for 2NF

Are any non-prime attributes determined by a subset of a candidate key?

Violation!

Step 5: Decompose with the violating FD

$R_3 = R_1 - \text{RHS of FD}$

$R_3 = \{\text{ActorID}, \text{Role}, \text{MovieID}\}$

$R_4 = \text{LHS} + \text{RHS of FD}$

$R_4 = \{\text{MovieID}, \text{MovieName}\}$

$R_2 = \{\text{ActorID}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}, \text{CountryCont}\}$

$R_3 = \{\text{ActorID}, \text{Role}, \text{MovieID}\}$

$R_4 = \{\text{MovieID}, \text{MovieName}\}$

$R_2 = \{\text{ActorID}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}, \text{CountryCont}\}$

$R_3 = \{\text{ActorID}, \text{Role}, \text{MovieID}\}$

$R_4 = \{\text{MovieID}, \text{MovieName}\}$

Step 1: 1NF

... generally good

Step 2: Write down our FDs:

$\text{ActorID} \rightarrow \text{ActorName}, \text{ActorDOB}, \text{CountryName}, \text{CountryCont}$

$\text{CountryName} \rightarrow \text{CountryCont}$

Step 3: Find candidate keys (minimal keys)

CKs: {ActorID}

Step 4: Check for 2NF

Are any non-prime attributes determined by a subset of a candidate key?

None found!

Step 5: Decompose with the violating FD (none)!

Step 6: Check for 3NF

Are any non-prime attributes determined by another non-prime attribute?

Violation!

Step 7: Decompose

$R_5 = R_2 - \text{RHS of FD}$

$R_5 = \{\text{ActorID}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}\}$

$R_6 = \text{LHS} + \text{RHS of FD}$

$R_6 = \{\text{CountryName}, \text{CountryCont}\}$

$R_3 = \{\text{ActorID}, \text{Role}, \text{MovieID}\}$

$R_4 = \{\text{MovieID}, \text{MovieName}\}$

$R_5 = \{\text{ActorID}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}\}$

$R_6 = \{\text{CountryName}, \text{CountryCont}\}$

$R_3 = \{\text{ActorID}, \text{Role}, \text{MovieID}\}$ ← **1NF + no non-trivial FDs,
must be in 3NF already**

$R_4 = \{\text{MovieID}, \text{MovieName}\}$

$R_5 = \{\text{ActorID}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}\}$ ← **1NF + singleton CK with no other
non-trivial FDs, must be in 3NF
already**

$R_6 = \{\text{CountryName}, \text{CountryCont}\}$

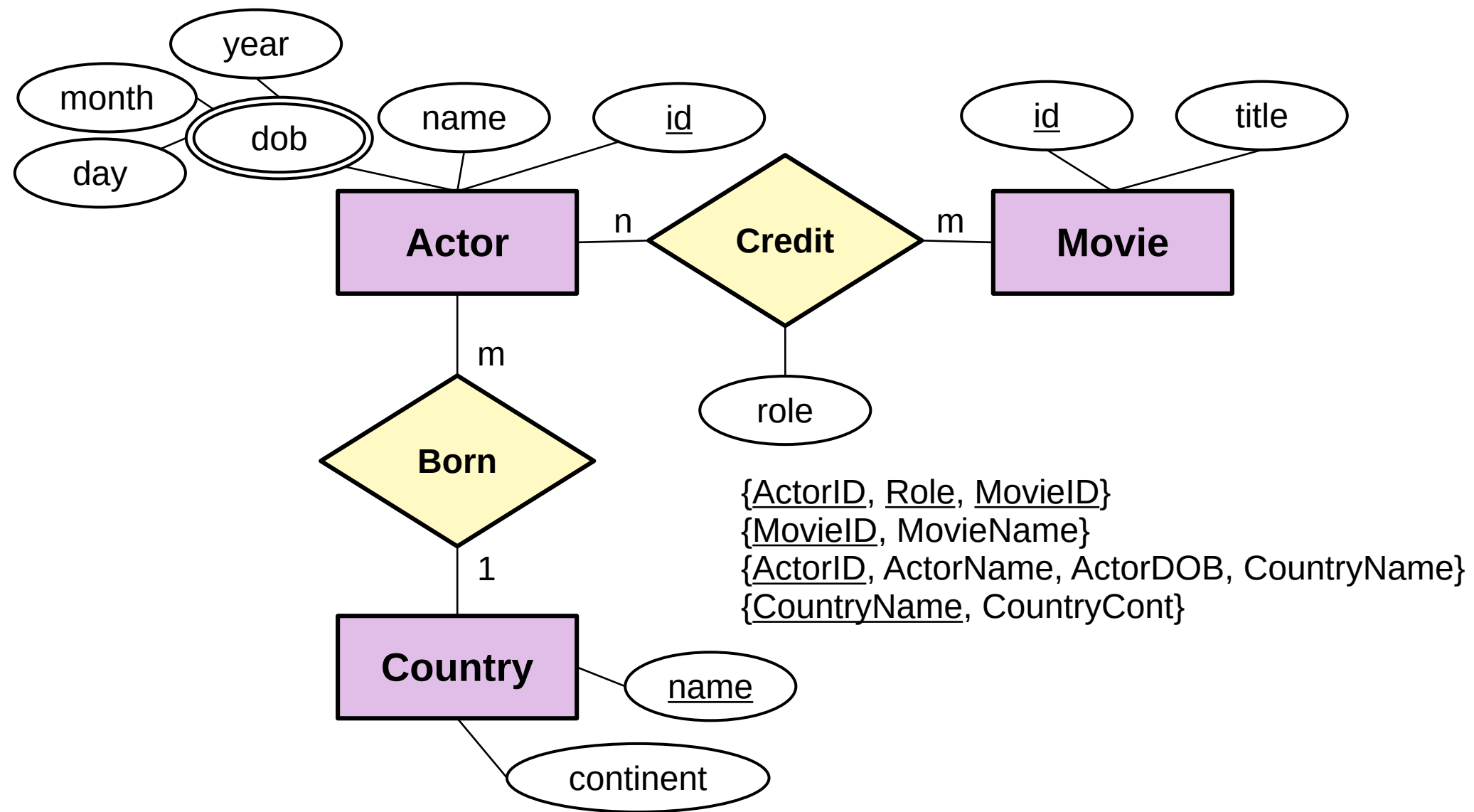
$R_3 = \{\underline{\text{ActorID}}, \text{Role}, \underline{\text{MovieID}}\}$

$R_4 = \{\underline{\text{MovieID}}, \text{MovieName}\}$

$R_5 = \{\underline{\text{ActorID}}, \text{ActorName}, \text{ActorDOB}, \text{CountryName}\}$

$R_6 = \{\underline{\text{CountryName}}, \text{CountryCont}\}$

Select a CK from each relation as the primary key. Underline it.



⋮ ▾ Lectures

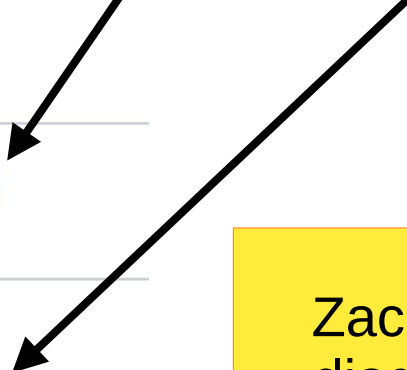
⋮ 📎 06 - Relational schema.pdf

⋮ 📎 (Reading) Textbook chapter on FDs and 3NF

⋮ 📎 (Reading) Textbook chapter on ER Diagrams

⋮ 📎 (Video) Knowledge & Entity Relationship Graphs ➞

Textbook chapters with
alternative treatments of FDs
and ER diagrams



Zack's lecture on ER
diagrams (extended)

